SOMAIYA VIDYAVIHAR UNIVERSITY

K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to the University of Mumbai

DEPARTMENT OF INFORMATION TECHNOLOGY

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Batch : SB4
Course Name and Code: Data Structures Lab (ITL302)
Semester: III (SYIT)
Academic Year: 2026-27 (Odd Semester)

EXPERIMENT 6

```
#include <stdio.h>
#include <stdlib.h>
#include <conio.h>

struct Node
{
int data;
struct Node *left;
struct Node *right;
};

struct Node* createNode(int data)
{
struct Node *newNode;

newNode = (struct Node*)malloc(sizeof(struct Node));

newNode->data = data;
newNode->left = NULL;
newNode->right = NULL;
```

```
return newNode;
}

void inorder(struct Node *root)
{
if (root != NULL)
{
inorder(root->left);
printf("%d ", root->data);
inorder(root->right);
}
}

void preorder(struct Node *root)
{
if (root != NULL)
{
printf("%d ", root->data);
preorder(root->left);
preorder(root->right);
}
```

```
}

void postorder(struct Node *root)
{
if (root != NULL)
{
postorder(root->left);
postorder(root->right);
printf("%d ", root->data);
}
}

void main()
{
struct Node *root;
clrscr();

root = createNode(1);

root->left = createNode(2);
root->right = createNode(3);
```

```
root->left = createNode(2);
root->right = createNode(3);

root->left->left = createNode(4);
root->left->right = createNode(5);

root->right->left = createNode(6);
root->right->right = createNode(7);

printf("\nInorder Traversal: ");
inorder(root);

printf("\nPreorder Traversal: ");
preorder(root);

printf("\nPostorder Traversal: ");
postorder(root);
getch();
}
```

OUTPUT

```
Inorder Traversal: 4 2 5 1 6 3 7
Preorder Traversal: 1 2 4 5 3 6 7
Postorder Traversal: 4 5 2 6 7 3 1 _
```